

The numerical range is a 2-spectral set

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Abstract

For an $n \times n$ complex matrix A , let $W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}$ be its numerical range. Here x^* denotes the conjugate transpose, $\|\cdot\|_2$ the Euclidean norm, and $\|\cdot\|$ the induced operator norm. For every complex polynomial p , we prove that

$$\|p(A)\| \leq 2 \max_{z \in W(A)} |p(z)|.$$

More generally, the same estimate holds for every function f holomorphic on a neighborhood of $W(A)$, with p replaced by f ; consequently, $W(A)$ is a 2-spectral set for A . This proves Crouzeix's conjecture, and the constant is optimal. The central ingredient is a mass-parameterized positive-real completion theorem relative to an auxiliary eigenbasis: if a normalized matrix-valued Carathéodory function completes $\frac{2}{m}(I - wT)^{-1}$ modulo the adjoint algebra, then $\|T\| \leq m$ for $m \geq 2$. The numerical-range double layer has mass two. The classical positive double-layer calculus supplies such completions for $f(B)$ whenever the auxiliary matrix B has simple spectrum; the eigenvalues of $f(B)$ may repeat. Sampling the associated Herglotz kernel at scaled conjugate diagonal entries and at the origin cancels the nonconstant completion term and leaves two ordered weighted Gramians; their first nonconstant terms yield the mass bound. Simple-spectrum approximation and canonical convex outer domains with supported oriented radial boundaries yield the general theorem. The mass formulation also provides a concrete scalar proof protocol for annular, multiply connected, operator-radius, and abstract spectral-constant problems.

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1. Introduction

Let $\mathbb{C}$ denote the complex field and, for $n \geq 1$, let $\mathbb{M}_n(\mathbb{C})$ denote the algebra of $n \times n$ complex matrices. For $A \in \mathbb{M}_n(\mathbb{C})$, its numerical range is

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}. \quad (1)$$

Here x^* denotes the conjugate transpose, $\sigma(A)$ denotes the spectrum, $\|x\|_2$ is the Euclidean norm, $\|X\|$ is the induced operator norm, and scalar moduli are denoted by $|\cdot|$. The set $W(A)$ is compact, the Toeplitz–Hausdorff theorem makes it convex, and $\sigma(A) \subset W(A)$. Functions holomorphic near $\sigma(A)$ are evaluated using the holomorphic functional calculus.

A compact set $K \subset \mathbb{C}$ containing $\sigma(A)$ is called a C -spectral set for A if

$$\|r(A)\| \leq C \max_{z \in K} |r(z)| \quad (2)$$

for every rational function r pole-free on K . The following finite-dimensional theorem states the customary polynomial form of Crouzeix's conjecture together with its holomorphic and spectral-set extensions.

Theorem 1.1 (Finite-dimensional form). *Let $n \geq 1$ and let $A \in \mathbb{M}_n(\mathbb{C})$. For every $p \in \mathbb{C}[z]$,*

$$\|p(A)\| \leq 2 \max_{z \in W(A)} |p(z)|. \quad (3)$$

More generally, if f is holomorphic on a neighborhood of $W(A)$, then

$$\|f(A)\| \leq 2 \max_{z \in W(A)} |f(z)|. \quad (4)$$

Consequently, $W(A)$ is a 2-spectral set for A .

The factor 2 is optimal: for

$$J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad p(z) = z, \quad (5)$$

one has $\max_{z \in W(J)} |z| = 1/2$ and $\|J\| = 1$.

For later use, write

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}.$$

We use I for the identity matrix of the size dictated by context, X^* for the adjoint, put $\operatorname{Re} X = (X + X^*)/2$, and let $\operatorname{alg}(T)$ denote the unital algebra generated by T . For Hermitian matrices X and Y , we write $X \preceq Y$ (equivalently, $Y \succeq X$) when $Y - X$ is positive semidefinite.

Crouzeix proved the estimate with constant 2 for 2×2 matrices and formulated the general conjecture in [1]. His subsequent dimension-free theorem gave the constant 11.08 [2]. After further structural and numerical work [3], Crouzeix and Palencia proved that $W(A)$ is a $(1 + \sqrt{2})$ -spectral set [4]. Their double-layer and Cauchy-transform argument was clarified in [5] and developed in several directions in [6,7].

The intervening literature established the conjectured constant for perturbed Jordan blocks and, more generally, scalar translates of cyclic weighted shifts, as well as for tridiagonal 3×3 matrices with elliptic numerical range centered at an eigenvalue [8–10]. Numerical and extremal investigations appear in [11,12], and a broad account of the problem and its variants is given in [13]. Subsequent work has treated Blaschke-product level sets, cyclicity reductions, compressions of shifts, further nilpotent classes, and quantitative bounds for KMS matrices [14–17]. An alternative proof for a class of weighted shifts appears in [18]. Across these cited reductions, the best dimension-free universal bound remained $1 + \sqrt{2}$.

1.1. The symmetrized calculus

The positive boundary measure used below originates in the convex-domain integral calculus of Delyon–Delyon [19]; see also [4,20]. If B has numerical range in a convex domain Ω with continuously differentiable boundary and f is holomorphic on a neighborhood of $\overline{\Omega}$ with $|f| \leq 1$ on $\overline{\Omega}$, this calculus produces a unital positive map Φ and a holomorphic Cauchy companion g_f such that

$$2\Phi(f) = f(B) + g_f(B)^*. \quad (6)$$

Thus positivity directly controls a symmetrized functional calculus, while the target estimate concerns $f(B)$. The Crouzeix–Palencia argument couples the two terms in (6); the example in [5] shows that its abstract norm estimate is sharp at $1 + \sqrt{2}$. The corresponding general uniform-algebra theorem has the same sharp constant; its additional unital case was proposed as a possible route

to 2 in [21]. Schwenninger and de Vries revisited the double-layer method [22]. Malman et al. proved that a configuration-constant refinement gives a strict improvement for every fixed numerical-range shape; they also constructed thin quadrilaterals for which the resulting bounds approach $1 + \sqrt{2}$ [23]. Separately, continuity and compactness give a constant $C_n < 1 + \sqrt{2}$ in every fixed matrix dimension n [24].

The additional information used here is algebraic. For the approximants constructed in Section 5, the companion in (6) belongs to the algebra generated by B^* , where the simple-spectrum matrix B supplies an eigenbasis that also diagonalizes $f(B)$. Treating $f(B)$ and the companion jointly preserves this coupling.

1.2. Auxiliary-basis de-symmetrization

Our numerical-range argument uses the mass-two specialization of the general completion theorem proved below and the whole Cayley family associated with (6). It has four steps.

1. Applying Φ to

$$\zeta \mapsto \frac{1 + wf(\zeta)}{1 - wf(\zeta)}, \quad w \in \mathbb{D},$$

produces a matrix-valued Carathéodory function H , meaning that $H(0) = I$ and $\operatorname{Re} H(w) \succeq 0$.

2. For a simple-spectrum auxiliary matrix B , put $T = f(B)$. The double-layer identity gives the positive-real completion

$$H(w) - (I - wT)^{-1} \in \operatorname{alg}(B^*), \quad w \in \mathbb{D}. \quad (7)$$

3. The eigenbasis of B diagonalizes $T = f(B)$ even when values of f on $\sigma(B)$ coincide, and it turns the adjoint-algebra defect in (7) into a diagonal analytic correction. Sampling the Herglotz kernel at half the conjugate diagonal entries of T , together with one vector sample at the origin, cancels the correction exactly; kernel positivity accommodates repeated sampling points.
4. The surviving positive-matrix inequality compares two weighted Gramians at scales τ and τ^2 , where $\tau = m^{-1}$. Their ordered difference and the first nonconstant Gramian term yield $\|T\| \leq m$; here $m = 2$.

The proof uses two limits: simple-spectrum matrices tend to A , followed by outer domains converging to $W(A)$. Together they yield the general case.

2. Positive-real kernels

Let $\mathcal{Q}$ be an $\mathbb{M}_n(\mathbb{C})$ -valued kernel on $\mathbb{D} \times \mathbb{D}$. We call $\mathcal{Q}$ positive if, for every finite choice $w_0, \dots, w_m \in \mathbb{D}$ and $\xi_0, \dots, \xi_m \in \mathbb{C}^n$,

$$\sum_{i,j=0}^m \xi_i^* \mathcal{Q}(w_i, w_j) \xi_j \geq 0. \quad (8)$$

We shall use the following standard matrix-valued Herglotz kernel fact. A direct proof is included to fix the normalization.

Lemma 2.1 (Matrix Herglotz kernel). *Let $F: \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ be analytic with $\operatorname{Re} F(w) \succeq 0$ for every $w \in \mathbb{D}$. Then*

$$\mathcal{L}_F(w, z) = \frac{F(w) + F(z)^*}{1 - w\bar{z}} \quad (9)$$

is a positive kernel.

Proof. Fix $w_0, \dots, w_m \in \mathbb{D}$ and $\xi_0, \dots, \xi_m \in \mathbb{C}^n$, and put

$$\eta(\zeta) = \sum_{j=0}^m \frac{\xi_j}{1 - \bar{w}_j \zeta}, \quad |\zeta| = 1.$$

For $0 \leq r < 1$, the scalar Cauchy formula, applied entrywise, gives

$$\frac{1}{2\pi} \int_0^{2\pi} \eta(e^{it})^* 2 \operatorname{Re} F(re^{it}) \eta(e^{it}) dt = \sum_{i,j=0}^m \xi_i^* \frac{F(rw_i) + F(rw_j)^*}{1 - w_i \bar{w}_j} \xi_j. \quad (10)$$

The integrand on the left is nonnegative, so the right side is nonnegative. Letting $r \uparrow 1$ and using the continuity of F at the finitely many points w_i gives (8) for $\mathcal{Q} = \mathcal{L}_F$. $\square$

3. The positive-real completion principle

The following mass-parameterized completion theorem is the core of the argument. It converts a positive-real completion constrained by the adjoint algebra of an auxiliary simple-spectrum matrix into a norm bound equal to the mass parameter.

Theorem 3.1 (Positive-real completion of mass m). *Let $2 \leq m < \infty$, and let*

$$B = S \operatorname{diag}(\beta_1, \dots, \beta_n) S^{-1}, \quad T = S \Lambda S^{-1}, \quad \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n), \quad (11)$$

where S is invertible, the numbers $\beta_1, \dots, \beta_n$ are distinct, and $|\lambda_i| \leq 1$ for every i . The numbers λ_i may repeat. Let $H: \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ be analytic and satisfy

$$H(0) = I, \quad \operatorname{Re} H(w) \succeq 0 \quad (w \in \mathbb{D}), \quad (12)$$

and

$$H(w) - \frac{2}{m}(I - wT)^{-1} \in \operatorname{alg}(B^*) \quad (w \in \mathbb{D}). \quad (13)$$

Then $\|T\| \leq m$.

Proof. Put $\tau = m^{-1}$, so $0 < \tau \leq 1/2$, and set $G = S^* S \succ 0$. Since the β_i are distinct, polynomial interpolation shows that

$$\operatorname{alg}(B^*) = \left\{ (S^{-1})^* D S^* : D \text{ is diagonal} \right\}. \quad (14)$$

By (14), the matrix

$$\Theta(w) = S^* (H(w) - 2\tau(I - wT)^{-1}) (S^*)^{-1}$$

is diagonal for every w . On setting $\tilde{H}(w) = S^* H(w) S$, we obtain

$$\tilde{H}(w) = 2\tau G (I - w\Lambda)^{-1} + \Theta(w) G. \quad (15)$$

Since $\tilde{H}(0) = G$, one has $\Theta(0) = (1 - 2\tau)I$. Write

$$\Theta(w) = (1 - 2\tau)I + \Psi(w), \quad \Psi(w) = \operatorname{diag}(\psi_1(w), \dots, \psi_n(w)),$$

where $\Psi(0) = 0$. Thus

$$\tilde{H}(w) = F(w) + \Psi(w)G, \quad F(w) = (1 - 2\tau)G + 2\tau G(I - w\Lambda)^{-1}. \quad (16)$$

Since $\operatorname{Re} \tilde{H}(w) = S^* (\operatorname{Re} H(w)) S$, the real part of $\tilde{H}(w)$ is positive semidefinite. Consequently, Lemma 2.1 applies to $\tilde{H}$.

Define Hermitian matrices $P, Q, Y \in \mathbb{M}_n(\mathbb{C})$ by

$$P_{ij} = \frac{G_{ij}}{1 - \tau^2 \bar{\lambda}_i \lambda_j}, \quad Q_{ij} = \frac{G_{ij}}{1 - \tau \bar{\lambda}_i \lambda_j}, \quad Y = Q - P. \quad (17)$$

Here $|\tau^2 \bar{\lambda}_i \lambda_j| \leq 1/4$ and $|\tau \bar{\lambda}_i \lambda_j| \leq 1/2$, so the denominators are nonzero. Since $G = G^*$, conjugate symmetry in the displayed formulas gives $P = P^*$ and $Q = Q^*$. Let $e_1, \dots, e_n$ be the standard basis of $\mathbb{C}^n$, and fix $u = (u_1, \dots, u_n)^T \in \mathbb{C}^n$. In the positive-kernel inequality use the n points and vectors

$$w_i = \tau \bar{\lambda}_i, \quad \xi_i = u_i e_i \quad (1 \leq i \leq n), \quad (18)$$

and add the point $w_0 = 0$ with vector

$$\xi_0 = v = -G^{-1}Pu. \quad (19)$$

The indexed sample remains valid when points coincide, including indices for which $\lambda_i = 0$. Its contribution from Ψ to the sampled kernel quadratic form is

$$2 \operatorname{Re} \sum_{i=1}^n \bar{u}_i \psi_i(w_i) \left((Gv)_i + \sum_{j=1}^n \frac{G_{ij} u_j}{1 - w_i \bar{w}_j} \right). \quad (20)$$

Because

$$1 - w_i \bar{w}_j = 1 - \tau^2 \bar{\lambda}_i \lambda_j,$$

the parenthesis in (20) is $(Gv + Pu)_i$, which vanishes by (19).

It remains to calculate the contribution of F . For $x = \bar{\lambda}_r \lambda_s$, the sample-sample entry is

$$2(1 - 2\tau)P_{rs} + \frac{4\tau G_{rs}}{(1 - \tau x)(1 - \tau^2 x)}. \quad (21)$$

The partial-fraction identity

$$\frac{1}{(1 - \tau x)(1 - \tau^2 x)} = \frac{1}{1 - \tau} \left(\frac{1}{1 - \tau x} - \frac{\tau}{1 - \tau^2 x} \right)$$

shows that the compressed sample-sample block is

$$M = 2P + \frac{4\tau}{1 - \tau} Y.$$

The sample-origin block is $R_0 = 2(1 - \tau)G + 2\tau Q$, and the origin-origin block is $2G$. After cancellation of the Ψ term, kernel positivity on the graph $v = -G^{-1}Pu$ gives

$$0 \preceq M - R_0 G^{-1} P - P G^{-1} R_0 + 2P G^{-1} P. \quad (22)$$

Using $Q = P + Y$ reduces this to

$$\frac{4\tau}{1 - \tau} Y - 2\tau(YG^{-1}P + PG^{-1}Y) + 2(1 - 2\tau)(PG^{-1}P - P) \succeq 0. \quad (23)$$

Balance the diagonalization by defining

$$\begin{aligned} \tilde{T} &= G^{1/2} \Lambda G^{-1/2}, & \hat{P} &= G^{-1/2} P G^{-1/2}, \\ \hat{Q} &= G^{-1/2} Q G^{-1/2}, & \hat{Y} &= \hat{Q} - \hat{P}. \end{aligned} \quad (24)$$

Expanding the scalar kernels in (17) into geometric series gives

$$\hat{P} = \sum_{k=0}^{\infty} \tau^{2k} \tilde{T}^{*k} \tilde{T}^k, \quad \hat{Q} = \sum_{k=0}^{\infty} \tau^k \tilde{T}^{*k} \tilde{T}^k. \quad (25)$$

Indeed, $\tilde{T}^k = G^{1/2} \Lambda^k G^{-1/2}$, so the sequence $(\tilde{T}^k)$ is bounded even when some $|\lambda_j| = 1$; the geometric weights give norm convergence. It follows that

$$\hat{P} \succeq I, \quad \hat{Y} = \sum_{k=1}^{\infty} (\tau^k - \tau^{2k}) \tilde{T}^{*k} \tilde{T}^k \succeq \frac{1-\tau}{\tau} (\hat{P} - I). \quad (26)$$

The last inequality follows termwise. Taking the congruence of (23) by $G^{-1/2}$ and dividing by 2τ yields

$$\frac{2}{1-\tau} \hat{Y} - \hat{Y} \hat{P} - \hat{P} \hat{Y} + \frac{1-2\tau}{\tau} (\hat{P}^2 - \hat{P}) \succeq 0. \quad (27)$$

Let e be a unit eigenvector of $\hat{P}$, say $\hat{P}e = \alpha e$, and put $y = e^* \hat{Y} e$. Then $\alpha \geq 1$ and (27) gives

$$0 \leq 2 \left(\frac{1}{1-\tau} - \alpha \right) y + \frac{1-2\tau}{\tau} \alpha (\alpha - 1). \quad (28)$$

If $\alpha \leq (1-\tau)^{-1}$, then $\alpha \leq 2$. Otherwise the coefficient of y is negative, while $y \geq (1-\tau)(\alpha-1)/\tau$ by (26). Because the coefficient is negative, this lower bound for y gives an upper bound for the right side of (28), namely

$$0 \leq \frac{\alpha-1}{\tau} (2-\alpha).$$

Thus $\alpha \leq 2$ in either case. Therefore

$$I \preceq \hat{P} \preceq 2I. \quad (29)$$

The $k = 1$ term of the same series now gives

$$\tau^2 \tilde{T}^{*} \tilde{T} \preceq \hat{P} - I \preceq I.$$

Thus $\|\tilde{T}\| \leq \tau^{-1} = m$. The polar decomposition $S = UG^{1/2}$ gives

$$T = U \tilde{T} U^*,$$

where U is unitary. Therefore $\|T\| \leq m$. $\square$

Remark 3.2. The sampling scale $\tau = m^{-1}$ simultaneously produces the weights τ and τ^2 in (25). Their ordered difference bounds $\hat{P}$, and the first nonconstant term then gives the coefficient m . The origin sample cancels the nonconstant diagonal correction exactly. When $m = 2$, the last term of (27) vanishes and the theorem reduces to the sharp completion used for numerical ranges. For $m > 2$, the positive constant part $\Theta(0) = (1 - 2/m)I$ supplies the additional Gramian term in (27). Distinct auxiliary eigenvalues identify $\text{alg}(B^*)$ with the diagonal algebra in the chosen basis, while the diagonal entries of T may repeat.

4. The double-layer completion

We now construct from the numerical range the mass-two function required by Theorem 3.1. The measure and companion transform used here are classical [2,4,19]. The full Cayley family yields the algebraic constraint (13) relative to the auxiliary matrix.

Section 5 uses the following canonical parallel domains. We say that a bounded convex domain Ω has a *supported oriented radial boundary* if, for some $c \in \Omega$, its boundary is traced once by

$$\gamma(t) = c + \rho(t)e^{it}, \quad 0 \leq t \leq 2\pi,$$

where ρ is positive, continuously differentiable, and 2π -periodic, and there are continuous 2π -periodic functions v and s such that $|v(t)| = 1$, $s(t) \geq 0$,

$$\gamma'(t) = iv(t)s(t), \quad \operatorname{Re}(\overline{v(t)}(\gamma(t) - z)) \geq 0 \quad (z \in \Omega). \quad (30)$$

Thus v is the outward unit normal, $s(t) dt$ is arclength, and the orientation is positive. Lemma 5.2 below supplies exactly this boundary for every canonical parallel domain.

Lemma 4.1 (Radial double-layer positive map). *Let $B \in \mathbb{M}_n(\mathbb{C})$, let $\Omega \subset \mathbb{C}$ be a bounded convex domain with a supported oriented radial boundary γ , and suppose $W(B) \subset \Omega$. There is a bounded complex-linear, unital, positive map on the compact parameter interval,*

$$\Phi_\gamma: C([0, 2\pi]) \longrightarrow \mathbb{M}_n(\mathbb{C}),$$

given by the double-layer density below.

Proof. Use the data in (30), and put

$$Z_t = \gamma(t)I - B, \quad \mathcal{R}_t = Z_t^{-1}.$$

This resolvent exists because $\sigma(B) \subset W(B) \subset \Omega$. Define the matrix-valued boundary density

$$D_B(t) = \frac{s(t)}{2\pi} \left(v(t)\mathcal{R}_t + \overline{v(t)}\mathcal{R}_t^* \right). \quad (31)$$

It is positive semidefinite. The support inequality in (30) gives

$$v(t)Z_t^* + \overline{v(t)}Z_t = 2\operatorname{Re}(\overline{v(t)}Z_t) \succeq 0. \quad (32)$$

Indeed, for every unit vector $x \in \mathbb{C}^n$, the quadratic form on the left is

$$2\operatorname{Re}(\overline{v(t)}(\gamma(t) - x^*Bx)) \geq 0,$$

because $x^*Bx \in W(B) \subset \Omega$. Congruence by $\mathcal{R}_t$ turns the left side of (32) into the matrix in parentheses in (31).

Since $d\gamma = iv(t)s(t) dt$, the matrix Cauchy formula gives

$$\frac{1}{2\pi} \int_0^{2\pi} v(t)\mathcal{R}_t s(t) dt = I. \quad (33)$$

Taking adjoints gives the same identity for the second layer. Hence

$$\int_0^{2\pi} D_B(t) dt = 2I. \quad (34)$$

Thus the raw positive layer has mass $2I$. For $\varphi \in C([0, 2\pi])$, define

$$\Phi_\gamma(\varphi) = \frac{1}{2} \int_0^{2\pi} \varphi(t) D_B(t) dt. \quad (35)$$

The factor $1/2$ normalizes this mass. The positivity of D_B and (34) make Φ_γ positive and unital, hence $*$ -preserving. It is bounded because

$$\|\Phi_\gamma(\varphi)\| \leq \frac{1}{2} \|\varphi\|_\infty \int_0^{2\pi} \|D_B(t)\| dt.$$

□

Proposition 4.2 (Radial Cayley completion). *Let Ω be a bounded convex domain with a supported oriented radial boundary, let $B \in \mathbb{M}_n(\mathbb{C})$ have simple spectrum, and suppose $W(B) \subset \Omega$. Let V be an open neighborhood of $\overline{\Omega}$, and let f be holomorphic on V and satisfy*

$$\max_{\zeta \in \overline{\Omega}} |f(\zeta)| \leq 1. \quad (36)$$

Put $T = f(B)$. Then there is an analytic $H: \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ satisfying

$$H(0) = I, \quad \operatorname{Re} H(w) \succeq 0 \quad (w \in \mathbb{D}), \quad (37)$$

and

$$H(w) - (I - wT)^{-1} \in \operatorname{alg}(B^*) \quad (w \in \mathbb{D}). \quad (38)$$

Proof. Choose

$$B = S \operatorname{diag}(\beta_1, \dots, \beta_n) S^{-1}.$$

Then $T = S \operatorname{diag}(f(\beta_1), \dots, f(\beta_n)) S^{-1}$, and (36) gives $|f(\beta_i)| \leq 1$ for every i . In particular, $\sigma(T) \subset \overline{\mathbb{D}}$.

For the fixed boundary data, put $\mathcal{R}_t = (\gamma(t)I - B)^{-1}$, and let Φ_γ be the bounded map from Lemma 4.1. For $w \in \mathbb{D}$, define

$$c_w(t) = \frac{1 + wf(\gamma(t))}{1 - wf(\gamma(t))} \quad (39)$$

and put

$$H(w) = \Phi_\gamma(c_w). \quad (40)$$

The expansion

$$c_w(t) = 1 + 2 \sum_{m=1}^{\infty} w^m f(\gamma(t))^m \quad (41)$$

converges locally uniformly in $C([0, 2\pi])$. The boundedness of Φ_γ therefore makes H analytic, and $H(0) = I$. Furthermore,

$$\operatorname{Re} c_w(t) = \frac{1 - |wf(\gamma(t))|^2}{|1 - wf(\gamma(t))|^2} \geq 0.$$

The positivity and $*$ -preservation of Φ_γ give $\operatorname{Re} H(w) = \Phi_\gamma(\operatorname{Re} c_w) \succeq 0$.

For every $m \geq 0$, the scalar Cauchy formula applied in the chosen eigenbasis gives exactly

$$\frac{1}{2\pi} \int_0^{2\pi} f(\gamma(t))^m v(t) \mathcal{R}_t s(t) dt = T^m. \quad (42)$$

Indeed, this is the diagonal collection of the scalar Cauchy formulas for f^m at the points β_i ; the neighborhood V contains the contour and its interior. Transporting the uniformly convergent series (41) through this first-layer integral gives

$$I + 2 \sum_{m=1}^{\infty} w^m T^m = (I + wT)(I - wT)^{-1}.$$

The matrix series converges because the powers of T are bounded in the displayed eigenbasis.

Define the companion matrix

$$C_f(w) = \frac{1}{2\pi} \int_0^{2\pi} \overline{c_w(t)} v(t) \mathcal{R}_t s(t) dt.$$

Splitting the density (31) into its two layers now gives

$$2H(w) = (I + wT)(I - wT)^{-1} + C_f(w)^*.$$

Each $\mathcal{R}_t$ belongs to $\text{alg}(B)$, and this finite-dimensional algebra is closed, so $C_f(w) \in \text{alg}(B)$. Since $(I + wT)(I - wT)^{-1} = 2(I - wT)^{-1} - I$, we conclude that

$$H(w) - (I - wT)^{-1} = \frac{1}{2}(C_f(w)^* - I) \in \text{alg}(B^*). \quad (43)$$

This proves (38). $\square$

Corollary 4.3 (Auxiliary sharp bound). *Under the hypotheses of Proposition 4.2, $\|f(B)\| \leq 2$.*

Proof. Write

$$B = S \text{diag}(\beta_1, \dots, \beta_n) S^{-1}.$$

Then

$$T = f(B) = S \text{diag}(f(\beta_1), \dots, f(\beta_n)) S^{-1}.$$

Since $\sigma(B) \subset \Omega$, the bound on f gives $|f(\beta_i)| \leq 1$ for every i . Apply Theorem 3.1 with $m = 2$ to these simultaneous diagonalizations and to the function H constructed in Proposition 4.2. The values $f(\beta_i)$ may repeat. $\square$

5. The finite-dimensional theorem

Successive approximations pass from the auxiliary setting to arbitrary matrices.

Lemma 5.1 (Simple spectrum and numerical range). *Matrices with n distinct eigenvalues are dense in $\mathbb{M}_n(\mathbb{C})$. Moreover, for $A, B \in \mathbb{M}_n(\mathbb{C})$,*

$$W(B) \subset W(A) + \{z \in \mathbb{C} : |z| \leq \|B - A\|\}. \quad (44)$$

Proof. The discriminant of the characteristic polynomial is a nonzero polynomial in the matrix entries, as is seen at $\text{diag}(1, \dots, n)$. The zero set of a nonzero complex polynomial has empty interior, so its nonvanishing set is dense in $\mathbb{M}_n(\mathbb{C})$, proving the first assertion. For a unit vector x ,

$$|x^*(B - A)x| \leq \|B - A\|,$$

which proves (44). $\square$

Lemma 5.2 (Parallel convex domains). *Let $K \subset \mathbb{C}$ be nonempty, compact, and convex. For $\delta > 0$, the set*

$$\Omega_\delta = \{z \in \mathbb{C} : \text{dist}(z, K) < \delta\}$$

is a bounded convex domain, $\overline{\Omega}_\delta = K + \delta\overline{\mathbb{D}}$, and Ω_δ has a supported oriented radial boundary.

Proof. The set Ω_δ is bounded and convex, and its closure is the Minkowski sum $K + \delta\overline{\mathbb{D}}$. Identify $\mathbb{C}$ with $\mathbb{R}^2$. The metric projection $\pi_K: \mathbb{C} \rightarrow K$ is unique and continuous. Put $d_K(z) = \text{dist}(z, K)$. For $z, h \in \mathbb{C}$, comparison first with $\pi_K(z)$ and then with $\pi_K(z + h)$ gives

$$\begin{aligned} 2 \text{Re}(\overline{z + h - \pi_K(z + h)} h) - |h|^2 &\leq d_K(z + h)^2 - d_K(z)^2 \\ &\leq 2 \text{Re}(\overline{z - \pi_K(z)} h) + |h|^2. \end{aligned}$$

The continuity of π_K now shows that d_K^2 is continuously differentiable, with real gradient $2(z - \pi_K(z))$. On the level set $d_K(z) = \delta$ this gradient has norm $2\delta > 0$. Since $\partial\Omega_\delta = \{d_K = \delta\}$, the implicit-function theorem shows that $\partial\Omega_\delta$ is continuously differentiable.

Choose $c \in K$. The ball $c + \delta\mathbb{D}$ lies in Ω_δ , so every ray from c meets the boundary once and transversely. Its radial function ρ is therefore positive, continuously differentiable, and periodic. With

$$\nu(t) = \frac{\gamma(t) - \pi_K(\gamma(t))}{\delta} \quad (0 \leq t \leq 2\pi)$$

as the outward unit normal along γ and $s(t) = |\gamma'(t)|$, convexity gives the support inequality in (30); positive orientation gives $\gamma'(t) = i\nu(t)s(t)$. These data form the required supported oriented radial boundary. $\square$

Proof of Theorem 1.1. We prove the holomorphic estimate (4); the polynomial estimate (3) follows by taking $f = p$. Put $K = W(A)$, and let U be an open neighborhood of K on which f is holomorphic. Choose $\varepsilon_0 > 0$ such that the closed ε_0 -neighborhood of K lies in U . For $0 < \varepsilon < \varepsilon_0$, let

$$\Omega_\varepsilon = \{z \in \mathbb{C} : \text{dist}(z, K) < \varepsilon\}.$$

Lemma 5.2 makes this a bounded convex domain with a supported oriented radial boundary. The open convex set $V = \Omega_{\varepsilon_0}$ satisfies $\overline{\Omega_\varepsilon} \subset V \subset U$.

By Lemma 5.1, choose simple-spectrum matrices $B_k \rightarrow A$. For fixed ε , all sufficiently large k satisfy $\|B_k - A\| < \varepsilon/2$, and then (44) gives

$$W(B_k) \subset K + (\varepsilon/2)\overline{\mathbb{D}} \subset \Omega_\varepsilon.$$

Set

$$m_\varepsilon = \max_{z \in \overline{\Omega_\varepsilon}} |f(z)|.$$

If $m_\varepsilon = 0$, then f vanishes on Ω_ε , so $f(A) = 0$. If $m_\varepsilon > 0$, apply Corollary 4.3 to B_k and f/m_ε to obtain

$$\|f(B_k)\| \leq 2m_\varepsilon.$$

Continuity of the holomorphic functional calculus, obtained from a fixed contour inside U , gives $f(B_k) \rightarrow f(A)$ and hence the same upper bound for $f(A)$. Thus, in either case,

$$\|f(A)\| \leq 2m_\varepsilon. \quad (45)$$

Put $m_0 = \max_{z \in K} |f(z)|$. Uniform continuity of f on the closed ε_0 -neighborhood of K gives a modulus of continuity ω_f there. Since every point of $\overline{\Omega_\varepsilon}$ lies within ε of K ,

$$0 \leq m_\varepsilon - m_0 \leq \omega_f(\varepsilon).$$

Letting $\varepsilon \downarrow 0$ in (45) proves (4).

Finally, a rational function pole-free on the compact set $W(A)$ is holomorphic on a neighborhood of $W(A)$. Applying (4) to such functions proves (2) with $K = W(A)$ and $C = 2$. $\square$

6. Consequences and scope

The finite-dimensional polynomial estimate passes to bounded Hilbert-space operators by the standard compression argument from [2]; compare the survey [13]. Polynomial approximation then gives the corresponding holomorphic and spectral-set forms.

Corollary 6.1 (Operator-level form). *Let $\mathcal{H} \neq \{0\}$ be a complex Hilbert space, with inner product linear in the first variable, and let $\mathcal{B}(\mathcal{H})$ denote its algebra of bounded operators. For $A \in \mathcal{B}(\mathcal{H})$, put*

$$W(A) = \{\langle Ax, x \rangle : x \in \mathcal{H}, \|x\| = 1\}, \quad K = \overline{W(A)}.$$

Then, for every $p \in \mathbb{C}[z]$,

$$\|p(A)\| \leq 2 \sup_{z \in W(A)} |p(z)|. \quad (46)$$

More generally, if f is holomorphic on a neighborhood of K , then

$$\|f(A)\| \leq 2 \max_{z \in K} |f(z)| = 2 \sup_{z \in W(A)} |f(z)|. \quad (47)$$

Consequently, K is a 2-spectral set for A .

Proof. We first prove (46). The assertion is immediate if $p = 0$. Otherwise, write $d = \deg p$. For a unit vector $x \in \mathcal{H}$, let

$$\mathcal{M}_x = \text{span}\{x, Ax, \dots, A^d x\}, \quad A_x = \Pi_x A|_{\mathcal{M}_x},$$

where Π_x is the orthogonal projection onto $\mathcal{M}_x$. Induction gives

$$A_x^k x = A^k x \quad (0 \leq k \leq d), \quad (48)$$

because $A^{k+1}x \in \mathcal{M}_x$ whenever $k < d$. Moreover, $W(A_x) \subset W(A)$. Theorem 1.1, applied on the finite-dimensional space $\mathcal{M}_x$, therefore gives

$$\|p(A)x\|_{\mathcal{H}} = \|p(A_x)x\|_{\mathcal{H}} \leq \|p(A_x)\| \leq 2 \sup_{z \in W(A)} |p(z)|.$$

Taking the supremum over unit vectors x proves (46).

The set K is compact and convex, and $\sigma(A) \subset K$. Let f be holomorphic on an open neighborhood U of K . Choose $\eta > 0$ such that

$$L_\eta = \{z \in \mathbb{C} : \text{dist}(z, K) \leq \eta\} \subset U.$$

The compact convex set L_η has connected complement, so polynomial Runge approximation gives polynomials p_j converging uniformly to f on L_η . Let

$$\Gamma = \partial\{z \in \mathbb{C} : \text{dist}(z, K) < \eta/2\},$$

oriented positively. Lemma 5.2 shows that Γ is a continuously differentiable contour enclosing $\sigma(A)$. The holomorphic functional calculus gives

$$p_j(A) - f(A) = \frac{1}{2\pi i} \int_{\Gamma} (p_j(z) - f(z))(zI - A)^{-1} dz,$$

and hence $p_j(A) \rightarrow f(A)$ in operator norm. Since

$$\sup_{z \in W(A)} |p_j(z)| = \max_{z \in K} |p_j(z)|,$$

passing to the limit in (46) proves (47). Every rational function pole-free on K is holomorphic on a neighborhood of K , so the spectral-set assertion follows. $\square$

The finite-dimensional holomorphic estimate also controls polynomial and rational approximation of matrix functions. Let $A \in \mathbb{M}_n(\mathbb{C})$, let f be holomorphic on a neighborhood of $W(A)$, and let s be either a polynomial or a rational function pole-free on $W(A)$. Then $f - s$ is holomorphic on a neighborhood of $W(A)$, and Theorem 1.1 gives

$$\|f(A) - s(A)\| \leq 2 \max_{z \in W(A)} |f(z) - s(z)|. \quad (49)$$

This is the sharp universal form of the numerical-range estimate that underlies applications to GMRES and rational Krylov methods; see [7,25,26] for the role of spectral sets in those settings.

7. Discussion: sharp consequences and open boundaries

The completion theorem separates a finite-dimensional algebraic engine from a geometric positive layer. We formulate the engine as a reusable criterion and give detailed applications with matching upper and lower constructions. The closing subsection lists related open sharpness questions.

7.1. The positive-layer criterion and the abstract problem

Let $\mathcal{A}$ be a uniform algebra on a compact space K , let $\theta: \mathcal{A} \rightarrow \mathbb{M}_n(\mathbb{C})$ be a continuous unital homomorphism, and suppose that there is a simple-spectrum matrix B such that $\theta(\mathcal{A}) \subset \text{alg}(B)$. The following criterion is the functional-calculus form of Theorem 3.1.

Proposition 7.1 (Positive-layer transfer). *Suppose $2 \leq m < \infty$ and that a bounded complex-linear map $\mathcal{P}: \mathcal{A} \rightarrow \mathbb{M}_n(\mathbb{C})$ satisfies*

$$\mathcal{P}(1) = mI, \quad \text{Re } h \geq 0 \text{ on } K \implies \text{Re } \mathcal{P}(h) \succeq 0, \quad \mathcal{P}(h) - \theta(h) \in \text{alg}(B^*) \quad (50)$$

for every $h \in \mathcal{A}$. Then $\|\theta\| \leq m$.

Proof. Fix $f \in \mathcal{A}$ with $\|f\|_K \leq 1$, put $T = \theta(f)$, and, for $w \in \mathbb{D}$, define

$$c_w = (1 + wf)(1 - wf)^{-1}, \quad H(w) = m^{-1}\mathcal{P}(c_w).$$

The diagonal entries of T in the auxiliary basis are contractive character values. Moreover, H is analytic, $H(0) = I$, and $\text{Re } H(w) \succeq 0$. Since

$$\theta(c_w) = 2(I - wT)^{-1} - I,$$

condition (50) gives

$$H(w) - \frac{2}{m}(I - wT)^{-1} = \frac{1}{m}(\mathcal{P}(c_w) - \theta(c_w) - I) \in \text{alg}(B^*).$$

Theorem 3.1 yields $\|\theta(f)\| \leq m$; taking the supremum over the unit ball of $\mathcal{A}$ proves the assertion. $\square$

The criterion asks for four verifiable items: positivity, total mass, an adjoint-algebra defect, and an auxiliary simple-spectrum model, together with approximation when needed. A complementary abstract approach based on extremal pairs and their representing extremal measures is developed in [27]. One sharp abstract consequence of the present criterion is the semisimple-range case of the problem posed in [21].

Corollary 7.2 (Semisimple abstract Crouzeix problem). *Let $\alpha: \mathcal{A} \rightarrow \mathcal{A}$ be a unital antilinear map and put*

$$\theta_\alpha(h) = \frac{1}{2}(\theta(h) + \theta(\alpha(h))^*).$$

If $\|\theta_\alpha\| \leq 1$ and the finite-dimensional commutative algebra $\theta(\mathcal{A})$ is semisimple, then $\|\theta\| \leq 2$. The coefficient 2 is best possible in this semisimple subclass.

Proof. For each unit vector $x \in \mathbb{C}^n$, the scalar functional $h \mapsto x^*\theta_\alpha(h)x$ is unital and contractive, hence extends to a state on $C(K)$. Thus θ_α preserves nonnegative real parts, and $\mathcal{P} = 2\theta_\alpha$ is a positive layer of mass two with

$$\mathcal{P}(h) - \theta(h) = \theta(\alpha(h))^*.$$

Simultaneously diagonalize the semisimple range and choose, in the same basis, an auxiliary matrix with distinct diagonal entries. Proposition 7.1 gives the upper bound.

For sharpness, take $0 < \varepsilon, t < 1$, put

$$T_\varepsilon = \begin{pmatrix} \varepsilon & 2\sqrt{1-\varepsilon^2} \\ 0 & -\varepsilon \end{pmatrix}, \quad T_{\varepsilon,t} = tT_\varepsilon, \quad \theta_{\varepsilon,t}(h) = h(T_{\varepsilon,t}), \quad \alpha(h) = \overline{h(0)}1$$

on the disk algebra $\mathcal{A} = A(\mathbb{D})$. The elliptic range of T_ε is contained in $\overline{\mathbb{D}}$, so $W(T_{\varepsilon,t}) \subset \mathbb{D}$. Apply Lemma 4.1 to the unit circle. The Cauchy calculation in the proof of Proposition 4.2 gives $2\Phi_\gamma(h \circ \gamma) = h(T_{\varepsilon,t}) + h(0)I$. Since Φ_γ is unital and positive, the disk double-layer map

$$h \longmapsto \Phi_\gamma(h \circ \gamma) = \frac{1}{2}(h(T_{\varepsilon,t}) + h(0)I) = (\theta_{\varepsilon,t})_\alpha(h)$$

is unital and contractive. Since $T_{\varepsilon,t}$ has distinct eigenvalues, $\theta_{\varepsilon,t}(\mathcal{A}) = \text{alg}(T_{\varepsilon,t})$ is semisimple. Finally,

$$\|\theta_{\varepsilon,t}\| \geq \|T_{\varepsilon,t}\| = t(1 + \sqrt{1-\varepsilon^2}).$$

Letting first $t \uparrow 1$ and then $\varepsilon \downarrow 0$ proves sharpness. $\square$

7.2. A sharp higher-mass disk problem

For $\varrho \geq 1$, let C_ϱ denote the class of bounded operators $A \in \mathcal{B}(\mathcal{H})$ for which there are a Hilbert space $\mathcal{K} \supset \mathcal{H}$ and a unitary $U \in \mathcal{B}(\mathcal{K})$ such that

$$A^k = \varrho P_{\mathcal{H}} U^k|_{\mathcal{H}}, \quad k \geq 1.$$

Here $P_{\mathcal{H}}$ is the orthogonal projection of $\mathcal{K}$ onto $\mathcal{H}$. For $\varrho \geq 2$, the associated positive layer gives the following exact constant.

Theorem 7.3 (C_ϱ disk calculus). *For each $\varrho \geq 2$, the least universal constant for matrices $A \in C_\varrho$ is ϱ :*

$$\|f(A)\| \leq \varrho \max_{|z| \leq 1} |f(z)| \quad (51)$$

for every rational function f pole-free on the closed disk. The same sharp estimate on arbitrary Hilbert spaces is the classical Okubo–Ando theorem [28].

Proof. First suppose that $\sigma(A) \subset \mathbb{D}$. On $|\zeta| = 1$ put

$$P_A(\zeta) = \frac{1}{\pi} \text{Re}[\zeta(\zeta I - A)^{-1}].$$

The Poisson-kernel characterization of C_ϱ in [29] gives $P_A(\zeta) \succeq (2 - \varrho)(2\pi)^{-1}I$. Hence

$$D_A(\zeta) = P_A(\zeta) - \frac{2 - \varrho}{2\pi} I \succeq 0.$$

The disk double-layer identity gives, for the disk algebra,

$$\int_{|\zeta|=1} D_A(\zeta) ds = \varrho I, \quad \int_{|\zeta|=1} h(\zeta) D_A(\zeta) ds = h(A) + (\varrho - 1)h(0)I. \quad (52)$$

The defect is scalar. Proposition 7.1 therefore proves (51) for simple-spectrum matrices with spectrum in $\mathbb{D}$.

Now let $A \in C_\varrho \cap \mathbb{M}_n(\mathbb{C})$ be arbitrary and define, for $z \in \mathbb{D}$,

$$Q_A(z) = \frac{1}{\pi} \operatorname{Re}(I - zA)^{-1} - \frac{2 - \varrho}{2\pi} I.$$

The same characterization gives $Q_A(z) \succeq 0$ and $Q_A(0) = \varrho(2\pi)^{-1}I$. Applying the scalar Harnack inequality to every quadratic form gives, for $0 < t < 1$ and $|z| \leq t$,

$$Q_A(z) \succeq \frac{1 - t}{1 + t} \frac{\varrho}{2\pi} I.$$

For $0 \leq s \leq 1$ and $|\zeta| = 1$, $D_{stA}(\zeta) = Q_A(st\bar{\zeta})$, so the full density family for tA is uniformly positive. Joint continuity in the matrix, s , and ζ shows that every sufficiently small perturbation of tA has positive densities for $0 \leq s \leq 1$ and spectrum in $\mathbb{D}$, hence belongs to C_ϱ . Choose simple-spectrum matrices $A_j \rightarrow tA$ with this property. The first part of the proof gives

$$\|f(A_j)\| \leq \varrho \max_{|z| \leq 1} |f(z)|.$$

Letting $A_j \rightarrow tA$ and then $t \uparrow 1$ proves the upper bound.

For optimality, put $J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $A = \varrho J$. The Sz.-Nagy–Foiaş criterion recorded in [29] becomes

$$I - 2(1 - \varrho^{-1}) \operatorname{Re}(\bar{z}A) - (2\varrho^{-1} - 1)|z|^2 A^* A \succeq 0.$$

For $A = \varrho J$, the determinant of the displayed 2×2 matrix is $1 - |z|^2$, and its lower-right entry is $1 + \varrho(\varrho - 2)|z|^2$. The matrix is therefore positive definite for $z \in \mathbb{D}$. Thus $A \in C_\varrho$, and the test $f(z) = z$ has ratio $\|A\| = \varrho$. This witness also proves sharpness of the Hilbert-space statement. $\square$

7.3. Sharp annular problems

Fix $R > 1$ and write

$$\mathbb{A}_R = \{z \in \mathbb{C} : R^{-1} < |z| < R\}, \quad \overline{\mathbb{A}}_R = \{z \in \mathbb{C} : R^{-1} \leq |z| \leq R\}.$$

Theorem 7.4 (Quantum annulus). *If A is an invertible Hilbert-space operator with $\|A\|, \|A^{-1}\| \leq R$, then*

$$\|f(A)\| \leq 2 \max_{z \in \overline{\mathbb{A}}_R} |f(z)| \quad (53)$$

for every rational f pole-free on $\overline{\mathbb{A}}_R$. The constant 2 is optimal for every fixed R , already among finite matrices.

Proof. Orient the outer circle counterclockwise and the inner circle clockwise, and parametrize each by oriented arclength $z = z(s)$. For a matrix B satisfying $\|B\|, \|B^{-1}\| < R$, define

$$\mu_B(z) = \frac{1}{2\pi i} \left[z'(s)(zI - B)^{-1} - \overline{z'(s)}(\bar{z}I - B^*)^{-1} \right], \quad D_B(z) = \mu_B(z) - \frac{1}{2\pi i} \frac{z'(s)}{z} I.$$

On the outer circle and on the inner circle, respectively, direct multiplication gives

$$\begin{aligned} 2\pi R D_B(z) &= (zI - B)^{-1} (R^2 I - BB^*) (\bar{z}I - B^*)^{-1} \succeq 0, \\ 2\pi R^{-1} D_B(z) &= (zI - B)^{-1} (BB^* - R^2 I) (\bar{z}I - B^*)^{-1} \succeq 0. \end{aligned} \quad (54)$$

The two winding numbers cancel, while the two Cauchy terms each have mass I . Thus D_B has mass $2I$, and the annular Cauchy identity gives

$$\int h(z) D_B(z) ds - h(B) = (\mathcal{C}_R \bar{h})(B)^* \in \operatorname{alg}(B^*),$$

where

$$(\mathcal{C}_R \bar{h})(\zeta) = \frac{1}{2\pi i} \int_{\partial \mathbb{A}_R} \frac{\overline{h(z)}}{z - \zeta} dz.$$

To obtain the matrix upper bound, let $A \in \mathbb{M}_n(\mathbb{C})$ satisfy $\|A\|, \|A^{-1}\| \leq R$. Choose $R' > R$ close enough to R that f is pole-free on $\overline{\mathbb{A}_{R'}}$, and choose simple-spectrum matrices $B_j \rightarrow A$ with $\|B_j\|, \|B_j^{-1}\| < R'$. Let $\mathcal{A}(\overline{\mathbb{A}_{R'}})$ be the algebra of functions continuous on $\overline{\mathbb{A}_{R'}}$ and holomorphic on $\mathbb{A}_{R'}$. Write $D_{B_j}^{(R')}$ for the displayed density with R' in place of R , and set $\theta_j(h) = h(B_j)$. Then θ_j is a continuous unital homomorphism into $\text{alg}(B_j)$, and the positive map

$$\mathcal{P}_j: \mathcal{A}(\overline{\mathbb{A}_{R'}}) \longrightarrow \mathbb{M}_n(\mathbb{C}), \quad \mathcal{P}_j(h) = \int_{\partial \mathbb{A}_{R'}} h(z) D_{B_j}^{(R')}(z) ds,$$

has mass $2I$ and, for every $h \in \mathcal{A}(\overline{\mathbb{A}_{R'}})$, satisfies

$$\mathcal{P}_j(h) - h(B_j) = (\mathcal{C}_{R'} \bar{h})(B_j)^* \in \text{alg}(B_j^*).$$

Proposition 7.1 yields

$$\|f(B_j)\| \leq 2 \max_{z \in \overline{\mathbb{A}_{R'}}} |f(z)|.$$

Letting $j \rightarrow \infty$ and then $R' \downarrow R$ proves the matrix estimate.

For an arbitrary Hilbert-space operator, the boundary-dilation theorem of McCullough and Pascoe [30] provides a Hilbert space $\mathcal{K} \supset \mathcal{H}$ and an invertible operator $A_\partial \in \mathcal{B}(\mathcal{K})$ such that

$$\begin{aligned} A^k &= P_{\mathcal{H}} A_\partial^k|_{\mathcal{H}} \quad (k \in \mathbb{Z}), \\ (R^2 + R^{-2})I - A_\partial^* A_\partial - (A_\partial^* A_\partial)^{-1} &= 0. \end{aligned} \tag{55}$$

If $Z = A_\partial^* A_\partial$, then $(Z - R^2 I)(Z - R^{-2} I) = 0$. Hence $Z = R^2 E + R^{-2}(I - E)$ for a projection E . Writing $A_\partial = U_\partial Z^{1/2}$, let u_0, p_0 be the canonical unitary and projection generators of $C(\mathbb{T}) *_{\mathbb{C}} \mathbb{C}^2$, where $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$, and put

$$j_R = u_0(Rp_0 + R^{-1}(1 - p_0)).$$

The universal property gives a $*$ -representation ρ_∂ with $\rho_\partial(u_0) = U_\partial$ and $\rho_\partial(p_0) = E$, and hence $\rho_\partial(j_R) = A_\partial$. Exel and Loring proved that this free product is residually finite-dimensional [31]. Every finite-dimensional $*$ -representation π sends j_R and j_R^{-1} to matrices of norm at most R . Thus, for every Laurent polynomial q ,

$$\|q(A)\| \leq \|q(A_\partial)\| \leq \|q(j_R)\| = \sup_{\pi \text{ finite dimensional}} \|q(\pi(j_R))\| \leq 2 \max_{\overline{\mathbb{A}_R}} |q|.$$

Choose $R'' > R$ so that f is pole-free on $\overline{\mathbb{A}_{R''}}$, and choose Laurent polynomials q_j converging uniformly to f there. The displayed estimate makes $q_j(A)$ Cauchy, and the holomorphic functional calculus identifies its limit with $f(A)$. This proves (53).

For sharpness, on $\mathbb{C}^{2n}$ let

$$W_n e_k = \omega_k e_{k+1} \pmod{2n}, \quad \omega_k = \begin{cases} R, & 0 \leq k < n, \\ R^{-1}, & n \leq k < 2n. \end{cases}$$

Then $\|W_n\| = \|W_n^{-1}\| = R$. For $g_n(z) = R^{-n}(z^n + z^{-n})$,

$$\max_{\overline{\mathbb{A}_R}} |g_n| = 1 + R^{-2n}, \quad W_n^n e_0 = W_n^{-n} e_0 = R^n e_n,$$

so $\|g_n(W_n)\| \geq 2$. The ratios tend to 2. This finite cyclic form complements the Hilbert-space lower-bound construction in [32]. $\square$

The same exact constant persists for the centered double-layer class. For an invertible matrix A and $z, w \in \mathbb{D}$, use the notation

$$F_{R,0}^A(z, w) = 2 \operatorname{Re} \left[(I - zA/R)^{-1} + (I - wA^{-1}/R)^{-1} \right] - 2I.$$

The class $\operatorname{DL}\mathbb{A}_R(0)$ consists of the matrices whose spectrum lies in $\mathbb{A}_R$ and for which this kernel is positive semidefinite on $\mathbb{D}^2$.

Corollary 7.5 (Centered double-layer annulus). *For every $R > 1$, the least universal finite-dimensional constant for the class $\operatorname{DL}\mathbb{A}_R(0)$ of Jury and Tsikalas is 2. Equivalently, every A in that class and every rational f pole-free on $\overline{\mathbb{A}}_R$ satisfy*

$$\|f(A)\| \leq 2 \max_{z \in \overline{\mathbb{A}}_R} |f(z)|.$$

Proof. Let $A \in \operatorname{DL}\mathbb{A}_R(0)$ and fix such an f . Choose $R' > R$ sufficiently close to R that f remains pole-free on $\overline{\mathbb{A}}_{R'}$. The defining kernel satisfies

$$F_{R',0}^A(z, w) = F_{R,0}^A((R/R')z, (R/R')w), \quad z, w \in \mathbb{D}.$$

For each unit vector x , the function

$$(z, w) \mapsto x^* F_{R,0}^A((R/R')z, (R/R')w) x$$

is nonnegative and pluriharmonic on a neighborhood of $\overline{\mathbb{D}}^2$, and its value at $(0, 0)$ is 2. The strong minimum principle makes it positive throughout $\overline{\mathbb{D}}^2$. Compactness of the unit sphere times $\overline{\mathbb{D}}^2$ yields a uniform positive lower bound. Hence all sufficiently small simple-spectrum perturbations A_j of A belong to $\operatorname{DL}\mathbb{A}_{R'}(0)$ and have spectrum in the open annulus. For each A_j , the normalized annular double-layer map constructed by Jury and Tsikalas [33] is unital and contractive. Its unnormalized form is a positive layer of mass two, and the double-layer identity leaves an adjoint Cauchy companion. Proposition 7.1 gives

$$\|f(A_j)\| \leq 2 \max_{z \in \overline{\mathbb{A}}_{R'}} |f(z)|.$$

Let $A_j \rightarrow A$ and then $R' \downarrow R$ to obtain the upper bound.

Every quantum-annulus matrix whose spectrum lies in $\mathbb{A}_R$ belongs to $\operatorname{DL}\mathbb{A}_R(0)$. Indeed, for a contraction X and $z \in \mathbb{D}$,

$$2 \operatorname{Re}(I - zX)^{-1} - I = (I - \bar{z}X^*)^{-1}(I - |z|^2 X^* X)(I - zX)^{-1} \succeq 0;$$

apply this identity to $X = A/R$ and $X = A^{-1}/R$ and add. The matrices W_n above provide the matching lower bound. $\square$

For $c \in \mathbb{C}$ and $0 < r < R$, let A be a bounded Hilbert-space operator such that $A - cI$ is invertible and

$$\|A - cI\| \leq R, \quad \|(A - cI)^{-1}\| \leq r^{-1}.$$

Then

$$\|f(A)\| \leq 2 \max_{r \leq |z-c| \leq R} |f(z)|$$

for every rational function f pole-free on the displayed annulus, and the factor 2 is optimal. Apply Theorem 7.4 to $(A - cI)/\sqrt{rR}$ with annular parameter $\sqrt{R/r}$, and transport the weighted-shift lower sequence.

7.4. Sharp geometric transfers

The numerical-range theorem also solves the scaled q -range problem by a rank-one similarity extraction. We include the argument because it is the step that determines the constant.

For $n \geq 2$, $A \in \mathbb{M}_n(\mathbb{C})$, and $0 < |q| \leq 1$, put

$$W_q(A) = \{y^*Ax : \|x\|_2 = \|y\|_2 = 1, y^*x = q\}, \quad \Omega_q(A) = q^{-1}W_q(A).$$

Let $r = |q|$ and $\varkappa = (1 + \sqrt{1 - r^2})/r$. A phase change in y gives $\Omega_q(A) = \Omega_r(A)$. For a unit vector x , put

$$d_A(x) = \left(\|Ax\|_2^2 - |x^*Ax|^2 \right)^{1/2}.$$

Tsing's disk formula [34] is

$$\Omega_r(A) = \bigcup_{\|x\|_2=1} \left\{ z : |z - x^*Ax| \leq \frac{\sqrt{1 - r^2}}{r} d_A(x) \right\}. \quad (56)$$

It implies both $W(A) \subset \Omega_r(A)$ and $\Omega_s(A) \subset \Omega_r(A)$ for $r \leq s \leq 1$. For a unit vector v , let P_v be the projection onto $\mathbb{C}v$ and set $S_v = I + (\varkappa - 1)P_v$.

Lemma 7.6 (Rank-one stretch and extraction). *For every unit v ,*

$$W(S_v A S_v^{-1}) \subset \Omega_r(A). \quad (57)$$

If $X \in \mathbb{M}_n(\mathbb{C})$, $\sigma(X) \subset \overline{\mathbb{D}}$, and

$$\|S_v \varphi_{\omega,a}(X) S_v^{-1}\| \leq K \quad (58)$$

for every unit v , $|\omega| = 1$, $0 \leq a < 1$, where $\varphi_{\omega,a}(z) = (\omega z - a)/(1 - a\omega z)$, then

$$\|X\| \leq \max\{1, K/\varkappa\}. \quad (59)$$

Proof. For a unit u , put $\ell = \|S_v^{-1}u\|_2$ and $m = \|S_v u\|_2$. Directly from the two eigenvalues $1, \varkappa$ of S_v ,

$$1 \leq \ell m \leq \frac{1}{2}(\varkappa + \varkappa^{-1}) = r^{-1}.$$

With $x = S_v^{-1}u/\ell$, $y = S_v u/m$, and $s = (\ell m)^{-1}$, one has $r \leq s \leq 1$, $y^*x = s$, and

$$u^* S_v A S_v^{-1} u = s^{-1} y^* A x \in \Omega_s(A) \subset \Omega_r(A).$$

This proves (57).

For the extraction, put $t = \|X\|$ and assume $t > 1$. Choose unit vectors x, y with $Xx = ty$, and choose ω so that $c = x^*(\omega y) = |x^*y| \in [0, 1)$. The strict inequality follows because $c = 1$ would give an eigenvalue of modulus $t > 1$. There is an $a \in [0, 1)$ satisfying

$$tc(1 + a^2) = a(1 + t^2); \quad (60)$$

take $a = 0$ when $c = 0$, and otherwise use the sign change between $a = 0$ and $a = 1$. Set

$$u = (I - a\omega X)x, \quad w = (\omega X - aI)x.$$

Since $I - a\omega X$ is invertible, $u \neq 0$; moreover, $\|\omega Xx\|_2 = t > a$, so $w \neq 0$. Then $\varphi_{\omega,a}(X)u = w$ and (60) gives $u \perp w$. It also gives

$$\|w\|_2^2 - t^2\|u\|_2^2 = atc(t^2 - 1)(1 - a^2) \geq 0.$$

Taking $v = w/\|w\|_2$, the stretch fixes u and multiplies w by $\varkappa$. Testing (58) on u yields $K \geq \varkappa t$, proving (59). $\square$

Theorem 7.7 (Scaled q -numerical ranges). *For each fixed q with $0 < |q| \leq 1$, the least constant valid uniformly over all $n \geq 2$ and $A \in \mathbb{M}_n(\mathbb{C})$ is*

$$C_q = \max \left\{ 1, \frac{2|q|}{1 + \sqrt{1 - |q|^2}} \right\}. \quad (61)$$

More precisely, for every $n \geq 2$, every $A \in \mathbb{M}_n(\mathbb{C})$, and every rational function f pole-free on $\Omega_q(A)$,

$$\|f(A)\| \leq C_q \max_{z \in \Omega_q(A)} |f(z)|.$$

Thus the polynomial conjecture proposed in [35] holds, together with its rational spectral-set form.

Proof. Let $M = \max_{\Omega_r(A)} |f|$, fix $\varepsilon > 0$, and put $X = f(A)/(M + \varepsilon)$. For every unit v , Lemma 7.6 puts $W(S_v A S_v^{-1})$ in $\Omega_r(A)$. Apply Theorem 1.1 to the composition

$$z \mapsto \varphi_{\omega, a}(f(z)/(M + \varepsilon)).$$

Similarity covariance gives (58) with $K = 2$; spectral mapping gives $\sigma(X) \subset \mathbb{D}$. The extraction conclusion and $2/\varkappa = 2r/(1 + \sqrt{1 - r^2})$, followed by $\varepsilon \downarrow 0$, prove the upper bound.

Constant functions force $C_q \geq 1$. For the second branch, use $J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Tsing's disk formula gives

$$\max_{z \in \Omega_r(J)} |z| = \frac{\varkappa}{2}.$$

For a unit vector $x = (\xi, \zeta)$ put $a_x = |\xi|$ and $b_x = |\zeta|$. Then $d_J(x) = b_x^2$, so every point in the disk indexed by x has modulus at most

$$a_x b_x + \frac{\varkappa^2 - 1}{2\varkappa} b_x^2 \leq \frac{\varkappa}{2}, \quad \frac{\varkappa}{2} - a_x b_x - \frac{\varkappa^2 - 1}{2\varkappa} b_x^2 = \frac{(\varkappa a_x - b_x)^2}{2\varkappa}.$$

Equality is attained at $a_x = (1 + \varkappa^2)^{-1/2}$ and $b_x = \varkappa(1 + \varkappa^2)^{-1/2}$. The identity polynomial has ratio $2/\varkappa$, proving sharpness of (61). $\square$

Two further exact consequences concern fixed ellipses and the Douglas–Paulsen norm. For $0 \leq \delta < 1$, define

$$K_\delta = \left\{ x + iy : \frac{x^2}{(1 + \delta)^2} + \frac{y^2}{(1 - \delta)^2} \leq 1 \right\}.$$

Theorem 7.8 (Fixed ellipses). *For every fixed $\delta \in [0, 1)$, every bounded Hilbert-space operator T with $\overline{W(T)} \subset K_\delta$, and every rational function f pole-free on K_δ ,*

$$\|f(T)\| \leq 2 \max_{z \in K_\delta} |f(z)|.$$

The least constant uniform in T and f is 2. The same conclusion holds for every nondegenerate ellipse after an affine change of variables.

Proof. The upper bound follows from Corollary 6.1. For $0 < \delta < 1$, let X_n be the cyclic weighted shift on $\mathbb{C}^{2n}$ with n consecutive weights 1 followed by n weights δ . Then

$$X_n^{2n} = \delta^n I, \quad \|X_n\| \leq 1, \quad \|X_n^{-1}\| \leq \delta^{-1}.$$

Put $T_n = X_n + \delta X_n^{-1}$. A numerical-range formula of Agler, Lykova, and Young [36] gives $W(T_n) \subset K_\delta$. Define

$$\mathcal{F}_0 = 2, \quad \mathcal{F}_1(z) = z, \quad \mathcal{F}_k(z) = z\mathcal{F}_{k-1}(z) - \delta\mathcal{F}_{k-2}(z) \quad (k \geq 2).$$

Induction gives $\mathcal{F}_k(z + \delta z^{-1}) = z^k + \delta^k z^{-k}$. Since $\partial K_\delta = \{e^{it} + \delta e^{-it} : 0 \leq t \leq 2\pi\}$, the maximum-modulus principle gives

$$\max_{K_\delta} |\mathcal{F}_n| = 1 + \delta^n, \quad \mathcal{F}_n(T_n) = 2X_n^n, \quad \|X_n^n\| = 1.$$

Thus the ratios are $2/(1 + \delta^n)$ and tend to 2. For $\delta = 0$, use $T = 2J$ and the identity polynomial, where J is the matrix in (5). $\square$

Let $\mathcal{R}_\delta = \{z : \delta < |z| < 1\}$, and for $\varphi \in H^\infty(\mathcal{R}_\delta)$ let $\|\varphi\|_{\text{dp}}$ be the supremum of $\|\varphi(X)\|$ over invertible X with $\|X\| \leq 1$ and $\|X^{-1}\| \leq \delta^{-1}$ and $\sigma(X) \subset \mathcal{R}_\delta$, as in [37].

Corollary 7.9 (Douglas–Paulsen norm). *For every $0 < \delta < 1$ and every $\varphi \in H^\infty(\mathcal{R}_\delta)$,*

$$\|\varphi\|_\infty \leq \|\varphi\|_{\text{dp}} \leq 2\|\varphi\|_\infty,$$

and the factor 2 is optimal.

Proof. Scalar operators give the first inequality. For the second, put $Y = \delta^{-1/2}X$ and apply Theorem 7.4 to $f(z) = \varphi(\delta^{1/2}z)$ when φ is holomorphic past the closed annulus. In general, write $\varphi(z) = \sum_{k \in \mathbb{Z}} a_k z^k$ and use the angular Abel means

$$\varphi_s(z) = \sum_{k \in \mathbb{Z}} s^{|k|} a_k z^k, \quad 0 < s < 1.$$

They are Poisson averages of rotations of φ , so $\|\varphi_s\|_\infty \leq \|\varphi\|_\infty$; they are holomorphic on $s\delta < |z| < s^{-1}$. Uniform Laurent approximation therefore permits application of Theorem 7.4, and the holomorphic functional calculus gives $\varphi_s(X) \rightarrow \varphi(X)$ as $s \uparrow 1$. This proves the upper bound for all of $H^\infty(\mathcal{R}_\delta)$. For the shifts X_n above, $X_n^{2n} = \delta^n I$ gives $\sigma(X_n) \subset \{z : |z| = \sqrt{\delta}\} \subset \mathcal{R}_\delta$. The functions $\chi_n(z) = z^n + \delta^n z^{-n}$ satisfy $\|\chi_n\|_\infty = 1 + \delta^n$ and $\|\chi_n(X_n)\| = 2$, proving sharpness. $\square$

Finally, a spherical disk is a closed disk, a closed half-plane, or the closed exterior of a disk in the Riemann sphere. Such a set D is a one-spectral set for A if $\sigma(A) \subset D$ and

$$\|f(A)\| \leq \sup_{z \in D} |f(z)|$$

for every rational function f with poles outside D . Let A be a bounded Hilbert-space operator and let D_1, D_2 be spherical disks, each a one-spectral set for A ; the general framework is developed in [38]. The Möbius reductions of Crouzeix and Greenbaum [7] send the disjoint-boundary case to a bounded-disk/exterior-disk pair and the crossing case to a disk-half-plane pair.

Corollary 7.10 (Two spherical disks). *The least constant valid uniformly for all such pairs is 2:*

$$\|f(A)\| \leq 2 \sup_{z \in D_1 \cap D_2} |f(z)|.$$

Here f is any rational function pole-free on $D_1 \cap D_2$. If the boundary circles are disjoint and $D_1 \cap D_2$ is doubly connected, the optimal constant for that fixed pair is 2.

Proof. Möbius covariance reduces the doubly connected case to a round annulus, already handled above. If the circles cross, a Möbius map with pole outside $D_1 \cap D_2$ converts the intersection into a

disk-half-plane intersection containing the numerical range of the transformed operator: the disk and half-plane one-spectral hypotheses give this containment through their affine and Cayley test functions. The intersection is convex, so Corollary 6.1 applies. Tangency follows by enlargement and a limit, and a redundant constraint has constant one. These cases give the upper bound. Transported cyclic annular shifts show that every doubly connected case has lower constant two, and hence that the uniform two-disk constant is exactly two. $\square$

7.5. Sharp formal transfers

Faithful representations give two additional exact consequences. If $\mathfrak{C}$ is a unital C^* -algebra, define the algebraic numerical range by

$$V_{\mathfrak{C}}(a) = \{\phi(a) : \phi \in \mathfrak{C}^{\vee}, \|\phi\| = \phi(1) = 1\}.$$

$\mathfrak{C}^{\vee}$ denotes the continuous dual of $\mathfrak{C}$. Then, for every rational function f pole-free on $V_{\mathfrak{C}}(a)$,

$$\|f(a)\| \leq 2 \max_{z \in V_{\mathfrak{C}}(a)} |f(z)|. \quad (62)$$

A faithful universal representation identifies $V_{\mathfrak{C}}(a)$ with the closure of the represented numerical range, so Corollary 6.1 proves the estimate. The matrix algebra example $a = 2J$ proves that the universal constant over all unital C^* -algebras is exactly two. Now let $\mathcal{H}$ be infinite-dimensional, let $\mathcal{K}(\mathcal{H})$ denote the compact operators, let $[A]$ be the coset of A in the Calkin algebra, and define $W_e(A) = V_{\mathcal{B}(\mathcal{H})/\mathcal{K}(\mathcal{H})}([A])$. Applying the same argument to this quotient gives, for every rational f pole-free on $W_e(A)$,

$$\|f([A])\| \leq 2 \max_{z \in W_e(A)} |f(z)|.$$

If $f(A)$ is defined, the left side is its essential norm. Infinite amplification of $2J$ proves sharpness; the same equality of universal constants is established in [39].

Operator radii preserve all of the sharp scalar information above. For $v \geq 1$, put

$$w_v(T) = \inf\{s > 0 : s^{-1}T \in C_v\}.$$

Let $\mathcal{U}$ be a unital Banach algebra satisfying

$$\|p(a)\| \leq \max_{|z| \leq 1} |p(z)| \quad (a \in \mathcal{U}, \|a\| \leq 1, p \in \mathbb{C}[z]),$$

and let $\Xi: \mathcal{U} \rightarrow \mathcal{B}(\mathcal{H})$ be a bounded unital homomorphism. Put

$$\|\Xi\|_{w_v} = \sup_{\|h\| \leq 1} w_v(\Xi(h)).$$

Badea, Crouzeix, and Klaja prove the exact identity [40] $\|\Xi\|_{w_v} = H_v(\|\Xi\|)$, where, for $s \geq 1$,

$$H_v(s) = \frac{s^2 + 1 + \sqrt{(s^2 + 1)^2 - 4v(2 - v)s^2}}{2vs}; \quad (63)$$

Each relevant functional-calculus domain satisfies the displayed von Neumann inequality: the compact cases use uniform algebras, and the Douglas–Paulsen calculus uses $H^\infty(\mathcal{R}_\delta)$. Consequently every sharp constant-two family above has exact w_v -constant

$$H_v(2) = \frac{5 + \sqrt{25 - 16v(2 - v)}}{4v}, \quad H_2(2) = \frac{5}{4}, \quad (64)$$

and the scaled q -range family has exact constant $H_v(C_q)$. Continuity and strict monotonicity of H_v transfer each lower sequence and prove equality. The C_q disk family similarly gives the exact two-parameter constant $H_v(q)$ for $q \geq 2$.

7.6. Related open problems

Several nearby sharp-constant questions remain open. The present mass criterion gives the finite-dimensional upper bounds $2 + c$ for the noncentered classes $\text{DLA}_R(c)$, $c > 0$, defined in [33], $N + 1$ for a disk with $N \geq 1$ pairwise disjoint circular holes lying strictly inside the outer disk under the Badea–Beckermann–Crouzeix norm and inverse-resolvent constraints, and 4 under the numerical-annulus conditions $w(A), w(A^{-1}) \leq R$, where $w(T) = \sup\{|z| : z \in W(T)\}$. Determining matching lower bounds is open. Definitions and prior bounds appear in [38,41,42]. Further open questions concern the optimal constant for an individual crossing lens, optimal fixed-domain constants for unbounded convex numerical ranges, for which aperture-dependent upper bounds are given in [43], polyannuli and intersections of three or more spherical disks, infinite-dimensional reduction for the full double-layer classes, the complete matrix-valued numerical-range problem [44], related abstract complete reductions [45], and the classification of extremizers suggested by the multiplicity phenomena in [12].

AI-assistance disclosure

OpenAI ChatGPT contributed the idea of sampling the matrix Herglotz kernel at scaled conjugate eigenvalues, with scale $1/2$ in the mass-two application, and adding the origin sample that cancels the nonconstant adjoint-algebra correction's contribution. This appears in the proof of Theorem 3.1, especially (18)–(20). It also assisted with exposition, bibliography, and typesetting. The human author assumes full responsibility for the correctness of all mathematical statements and for the integrity and accuracy of all citations.

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